

Science

Quarter 3 – Module 2

Gravity

**S L M**

SELF-LEARNING MODULE

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Science

Quarter 3 – Module 2

Gravity

Introductory Message

For the facilitator:

Welcome to the Science 6 Self-Learning Module on Gravity.

This module was collaboratively designed, developed and reviewed by educators both from public and private institutions to assist you, the teacher or facilitator in helping the learners meet the standards set by the K to 12 Curriculum while overcoming their personal, social, and economic constraints in schooling.

This learning resource hopes to engage the learners into guided and independent learning activities at their own pace and time. Furthermore, this also aims to help learners acquire the needed 21st century skills while taking into consideration their needs and circumstances.

In addition to the material in the main text, you will also see this box in the body of the module:



Notes to the Teacher

This contains helpful tips or strategies that will help you in guiding the learners.

As a facilitator, you are expected to orient the learners on how to use this module. You also need to keep track of the learners' progress while allowing them to manage their own learning. Furthermore, you are expected to encourage and assist the learners as they do the tasks included in the module.

For the learner:

Welcome to the Science 6 Self-Learning Module on Gravity.

This module was designed to provide you with fun and meaningful opportunities for guided and independent learning at your own pace and time. You will be enabled to process the contents of the learning resource while being an active learner.

CONTENT STANDARD

The learners demonstrate understanding of how friction affects movement of objects.

PERFORMANCE STANDARD

The learners should be able to produce an advertisement demonstrates road safety.

LEARNING COMPETENCY

The learners should be able to infer how gravity affects movements of different objects. **(S6FE - IIIa - c - 1)**.

LEARNING OBJECTIVES:

Cognitive:

- Define gravity.
- Describe how gravity works.
- Explain how to overcome gravity.

Affective:

- Show enthusiasm and creativity in performing activities

Psychomotor:

- Perform simple individual activities that demonstrate gravity.

INTRODUCTION

Hello, my dear pupils welcome to your second Self-Directed Learning Module in Science 6 for the Third Quarter. This Learning Resource of **Gravity** is developed for you to enjoy while learning. There are various activities and check-up tests you would like to require. This module is good for one week. Take note of the instructions as you journey through this module to possess a solid understanding of what you're expected to accomplish. Carefully read and follow the instructions to hit what you're expected to try to.

Answer keys are provided which you'll be able to seek advice from when checking your answers. Record your score at the Pupil's Progress Chart and inform your teacher on your development. Feel free to use this material and if progress isn't evident, you'll be able to make use of it again until such the required passing mark is achieved.

PRE-TEST

A. Write the letter of your best answer on the space provided before the number.

_____ 1. Where is the gravitational force directed as you stand on the floor?

- A. upward
- B. downward
- C. Forward
- D. backward

_____ 2. Which of the following equals the force exerted by the earth on you?

- A. your mass
- B. your weight
- C. half your mass
- D. half your weight

_____ 3. How distance between two objects affects the pull of gravity?

- A. The lesser the distance, the lesser the gravitational pull between them
- B. The greater the distance, the greater the gravitational pull between them.
- C. The greater the distance, the lesser the gravitational pull between them.
- D. None of these, gravitational pull between them depends on their weights.

_____ 4. Which is true about mass and gravity?

- A. The lesser the mass of the object, the greater the pull of gravity.
- B. The greater the mass of the object, the lesser the pull of gravity.
- C. The greater the mass of the object, the greater the pull of gravity.
- D. None of these, gravity and mass are not related.

_____ 5. Where is the gravitational force directed when you are holding a glass?

- A. upward
- B. downward
- C. into the glass
- D. away from the glass

_____ 6. If the mass of the earth is doubled, how much increase would the weight of the objects on earth be?

- A. halved
- B. a surface sloped uphill
- C. doubled
- D. remains the same

B. Write TRUE if the statement is correct and change the underlined word/s if it is wrong.

- _____ 7. Gravity is a contact force that acts between two separate objects.
- _____ 8. Gravity is the force exerted by the Earth which is equivalent to the mass of an object.
- _____ 9. Gravity on the Moon is less than the gravity on the Earth.
- _____ 10. The principle of gravity was discovered by Sir Isaac Newton.

REVIEW OF THE PREVIOUS MODULE

Forces are pushes or pulls that can change the motion, or movement, of objects. **Friction** and **Gravity** are two examples of force.

A. **Directions:** Write **True** if the statement is correct and **False** if the statement is incorrect.

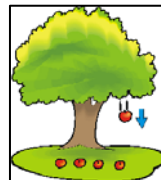
- _____ 1. Friction is contact force.
- _____ 2. Friction keeps object from sliding off surfaces.
- _____ 3. Friction is the force that opposes the pull of gravity on objects.
- _____ 4. Friction keeps anything from going up.
- _____ 5. The greater the friction the harder for the object to slip off.

B. Which of the following shows the presence of friction or gravity? Put a check✓ on the box if there's friction, and X if the picture illustrates the force of gravity.

1. ☐



2. ☐



3. ☐



4. ☐



5. ☐



PRESENTATION OF THE NEW MODULE

The moon and other planets maintain their orbits and do not collide. Things placed on tables, floors, and other surfaces do not simply float away.

Let's start with these activities to see how gravity functions in keeping things in order.

ACTIVITY

Activity No. 1 – Try to jump upward. Observe your body if it goes downward after jumping. Try to throw things upward, like a pencil, a ball pen, a book, etc. Observe them.

Activity No. 2 - Hold a ball and a coin. Put them in your left palm. Throw them up. Observe the objects.

Activity No. 3 - Drop a sheet of paper and a crumpled paper at the same time. Observe the objects.

ANALYSIS

Gravity is an attractive force that connects all matter together, and anything you can physically touch.

Guide questions:

1. What happen to the things you tried to throw upward?

2. What happen when you throw the ball and the coin?

3. Which of them reached the floor first? Why do you think so?

4. What happens when you dropped a crumpled sheet of paper and a sheet of paper?

5. Which of the papers was the first to hit the floor? Why?

6. What causes objects to fall?



Photo credit: Cyberscience 6, p. 188

Anything that is thrown upward falls down. This is because of gravity.6. _____ is the force that pulls object towards the ground. When an object falls, it moves faster and faster towards the ground. Falling objects are examples of accelerated motion due to the force of gravity.

In your activity, the ball and the coin that you have thrown upward fell to the ground at the same time. Gravity, according to Galileo, causes all objects to accelerate at the same rate. He also discovered that gravity accelerates all forces in the same way.

But, why is that a sheet of paper falls more slowly than a crumpled one?

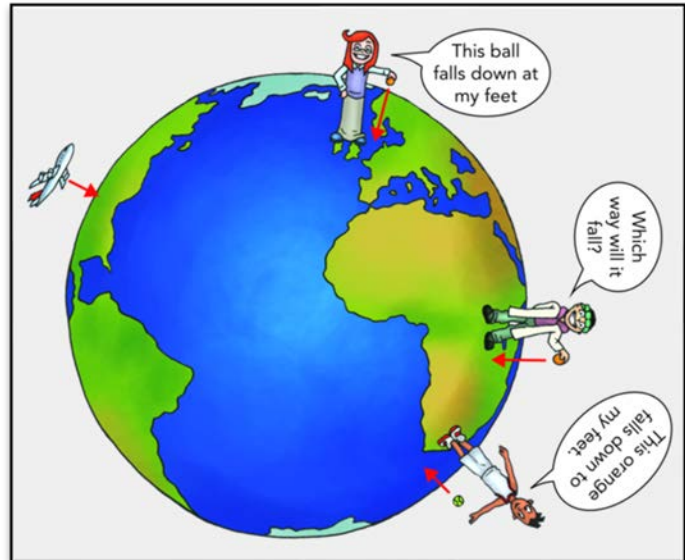
Your experiment with falling paper illustrates how air resistance impacts the movement of falling objects. Because of 7. _____, the sheet of paper falls slowly. The crumpled paper travels quicker because it has a smaller surface area that is not distorted by air. **The air resistance that affects the movement of a falling object increases as the surface area of the object increases.**

The idea of using a parachute is similar. Because of _____, the parachute falls slowly, ensuring that the individual using it can land safely..

ABSTRACTION

Anything that rises must eventually fall. Things do not leap off the surface of the earth. This is due to Earth's powerful attractive influence on you and all around you.

Gravity is the term for this phenomenon. This illustration depicts how gravity affects our world and everything on it..



<https://i.pinimg.com/originals/4d/bb/fc/4dbbfcba6177b92f887649a1c32f86d7.png>

Gravity and Mass

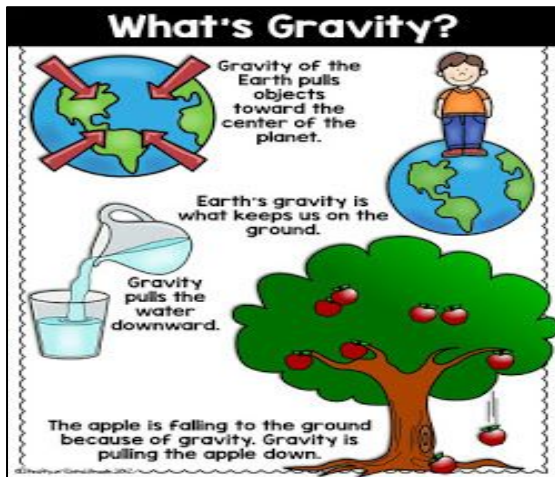
Sir Isaac Newton, a physicist, discovered the theory of gravity. All in the world, big or small, exerts gravity on other objects, according to this theory. The greater the mass of something, the greater the gravitational force. Bigger objects, such as planets and stars, have a greater gravitational force. Since the planet is huge, everything on it is highly attracted to it.



<https://clarkscience8.weebly.com/friction--gravity.html>

Gravity and Weight

Sir Isaac Newton, a physicist, discovered the theory of gravity. All in the world, big or small, exerts gravity on other objects, according to this theory. The greater the mass of something, the greater the gravitational force. Bigger objects, such as planets and stars, have a greater gravitational force. Since the planet is huge, everything on it is highly attracted to it.



On the other hand, since the Moon is smaller, has less mass, and has a weaker gravitational force than Earth, you will lose a lot of weight if you go there.

<https://i.pinimg.com/474x/fb/66/df/fb66dfb74529af1eae4f41671fd13889.jpg>

Gravity and Distance

Gravity is a **noncontact** force. It acts between objects even if they are separated and far from each other. The farther two objects are from each other, the smaller the gravitational pull will be between them.

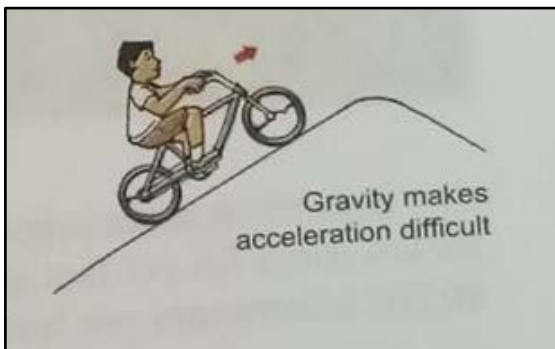


https://i.ytimg.com/vi/W55qMEF1a_E/hqdefault.jpg



https://www.theschoolrun.com/sites/theschoolrun.com/files/istock_gravity.jpg

Astronauts from space stations and spacecraft can float around and do the space walk. They are not weightless. They still have weight, but since they are far from Earth, it is less than when they are on the ground.

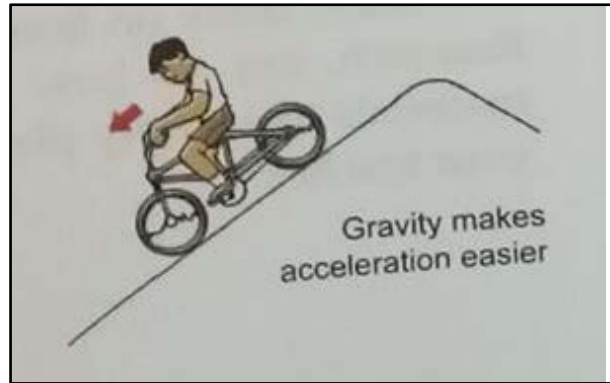


Have you ever ridden a bicycle up a mountain or a hill? Describe your pedaling technique. Why is it so difficult for you to accelerate as you ascend? When going downhill, compare the acceleration. Since you're going against gravity, going upward is difficult.

Gravity is dragging you down, and you'll need to expend a lot of effort to ride the bicycle hard enough to overcome gravity and get going.

However, since gravity is dragging you downhill, it will be easy to accelerate. Gravity is dragging you down. You don't have to apply any energy to travel because the bike's wheels roll automatically due to gravity's pull.

Photo credits: Cyberscience 6, p. 189



Escape Velocity

Space exploration demonstrates the connection between gravity on Earth and gravity in space. Because of gravity, you discovered that if you throw anything upward, it would fall. So, how do scientists launch spacecraft that aren't influenced by Earth's gravity? Scientists discovered that the gravitational force varies depending on the distance from Earth. A rocket must take off quickly enough to withstand gravity.

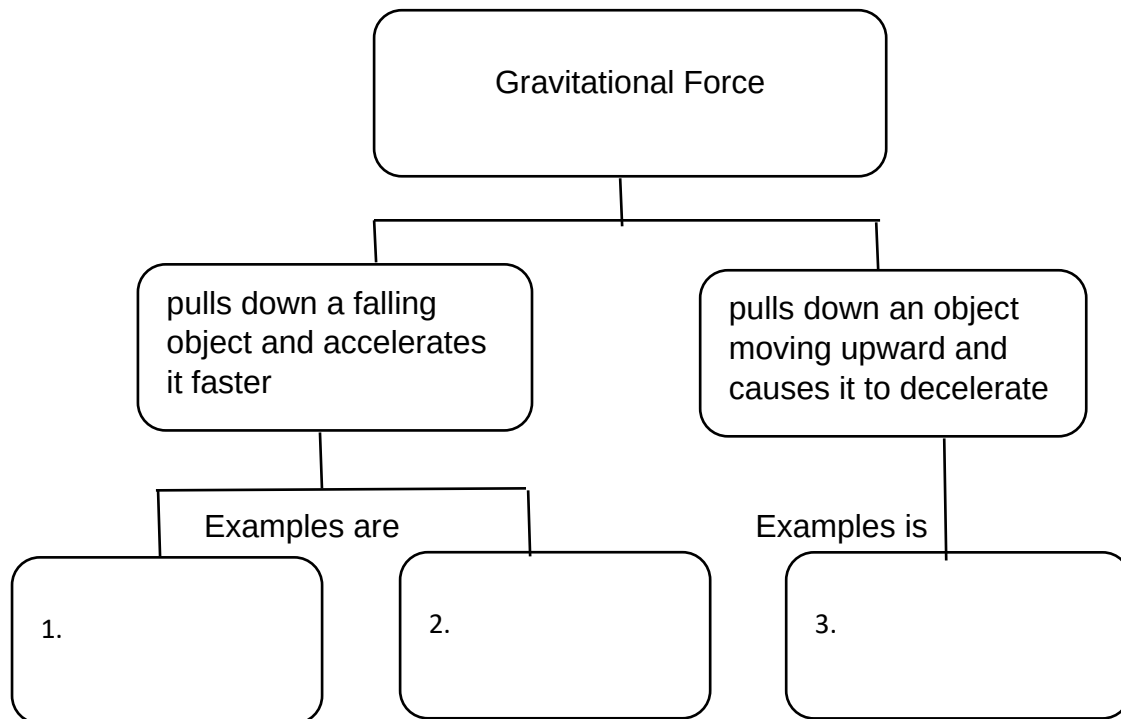
Scientists discovered that the gravitational force varies depending on the distance from Earth. A rocket must take off quickly enough to withstand gravity. To escape Earth's gravity, it must accelerate to about 40,200 kilometers per hour. This is known as **escape velocity**.



<https://i.pinimg.com/originals/bc/93/fc/bc93fcfeb94b453675f13affccd4de63.png>

APPLICATION

Complete the concept map below. Give examples of activities that show effects of gravitational force.



Complete the table below.

What would happen if gravity didn't exist?	
1.	
2.	
3.	
4.	
5.	

POST-TEST

A. Write the letter of your best answer on the space provided before the number.

_____ 1. If the mass of the earth is doubled, how much increase would the weight of the objects on earth be?

- A. halved
- B. a surface sloped uphill
- C. doubled
- D. remains the same

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- C. The greater the distance, the lesser the gravitational pull between them.
- D. None of these, gravitational pull between them depends on their weights.

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- B. half your mass
- C. your weight
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_____ 4. Which is true about mass and gravity?

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- D. away from the glass

_____ 6. Where is the gravitational force directed as you stand on the floor?

- A. upward
- B. downward
- C. forward
- D. backward

B. Write TRUE if the statement is correct and change the underlined word/s if it is wrong.

_____ 7. The principle of gravity was discovered by Sir Isaac Newton.

_____ 8. Gravity is the force exerted by the Earth which is equivalent to the mass of an object.

_____ 9. Gravity on the Moon is less than the gravity on the Earth.

_____ 10. Gravity is a contact force that acts between two separate objects.

ANSWER KEY

- Pre-Test
1. B
 2. B
 3. C
 4. C
 5. B
 6. C
 7. noncontact force
 8. weight
 9. TRUE
 10. TRUE
- Review of previous Lesson
1. True
 2. False
 3. False
 4. False
 5. True
- Analysis
1. fell down
 2. they fell at the same time
- Application
- (Answers may vary)
1. C
 2. C
 3. B
 4. C
 5. B
 6. B
 7. TRUE
 8. WEIGHT
 9. TRUE
 10. NONCONTACT FORCE
- Post Test
1. C
 2. C
 3. B
 4. C
 5. B
 6. B
 7. TRUE
 8. WEIGHT
 9. TRUE
 10. NONCONTACT FORCE
- area
3. fell down
 4. the flat sheet of paper because it has greater surface
 5. Gravity
 6. Gravity
 7. Air Resistance
 8. Air Resistance

REFERENCES

Book:

Adauro, M., & Valencia, N. (2016). *Cyberscience 6*. Rex Book Store.

Sarte, Evelyn, et al. (2016). *Science Beyond Borders 6*. Vibal Group, Inc.

Internet Sources:

<https://owlcation.com/stem/how-to-do-a-friction-science-experiment>

<https://www.ducksters.com/science/friction.php>

<https://www.dkfindout.com/us/science/forces-and-motion/friction/>

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